Collaborative financial fraud and money laundering detection

"How can banks detect fraud and money laundering collaboratively across institutions without sharing raw customer data? Compare federated learning, private set intersection and secure multi-party computation, and list the key papers." *Generated by LeapSpace*

Banks can collaborate on fraud and money-laundering detection without sharing raw customer data by combining **federated learning (FL)** for joint model training, **private set intersection (PSI)** for privacy-preserving record matching, and **secure multi-party computation (SMPC/MPC)** for privacy-preserving joint computation on encrypted or secret-shared data. The trade-off is simple: **FL is best for learning shared detection models**, **PSI is best for aligning identifiers or finding overlapping records**, and **SMPC is best for computing sensitive functions jointly without exposing inputs**. [88] [89] [90] [91] [92] [93] [94] [95]

How the three approaches differ

Approach	What it does	Strengths	Main limitations
Federated learning	Trains a shared fraud/AML model across banks while keeping data local; only model updates move between parties. [88] [89] [96] [97]	Good for cross-bank prediction, supports large-scale collaborative modeling, and can work with privacy add-ons like secure aggregation, DP, blockchain, or encryption. [96] [98] [99] [100] [101] [102]	Sensitive to **non-IID data**, communication overhead, poisoning/inference risks, and performance drops when clients are highly heterogeneous or small. [103] [104] [105] [106] [107]
Private set intersection	Lets banks find overlapping customers/accounts or matched identifiers without revealing non-matching records. [91] [92] [93]	Useful for **entity alignment**, duplicate detection, watchlist matching, and joining records across institutions before modeling. [91] [92] [108]	Exact PSI can be expensive at scale; large sets make computation and communication heavy, especially when the overlap is sparse. [93] [109]
SMPC / MPC	Splits inputs into secret shares so banks can compute a result jointly without revealing each party's raw values. [95] [108] [110] [111]	Strong privacy for aggregation, PageRank-style graph computations, secure scoring, and secure model updates. [94] [95] [98] [99] [112]	Usually more complex and resource-intensive than plain FL, with higher computation and communication cost. [93] [110] [111] [113]

What each is best for

- **Federated learning** is the main tool when the goal is to **build a shared fraud or AML detector** across banks while data stays local. Several studies report strong results in financial fraud settings, including cross-bank fraud detection, privacy-preserving AML detection, and secure multi-branch monitoring. [96] [98] [99] [100] [114] [115]
- **PSI** is most useful when banks first need to **identify the same customer, account, or entity across institutions** without revealing the full datasets. The evidence also shows that PSI becomes a bottleneck at very large scale, which is why preprocessing or threshold variants are proposed. [91] [92] [93] [109]
- **SMPC/MPC** is most useful for **joint computations over sensitive graph or transaction data**, such as secure aggregation, secure PageRank, or AML score computation. The evidence repeatedly frames MPC as the privacy layer that prevents gradient leakage or reveals only the final result. [94] [95] [98] [108] [111]

Practical reading of the trade-offs

- If banks want a **shared detector**, start with **FL** and add **secure aggregation + DP** or **encryption** for protection. [96] [98] [99] [114]
- If banks must **match entities first**, use **PSI** as the alignment layer; at scale, use optimizations because naïve PSI is costly. [91] [92] [93] [109]
- If banks need **joint statistics or graph features without disclosure**, use **SMPC/MPC**, especially for aggregation or graph computations. [94] [95] [98] [112]

Key papers

Federated learning

- **Privacy-Preserving Federated Learning for Detecting Fraudulent Financial Transactions in Japanese Banks** (2022) — real-bank demo; detected fraud cases that a single bank could not, and identified criminals' accounts before freezing. [116]
- **Credit card fraud detection based on federated graph learning** (2024) — combined FL and GNNs; improved recall and AUC over baselines. [117]
- **Fed-RD: Privacy-Preserving Federated Learning for Financial Crime Detection** (2024) — used differential privacy and SMPC; achieved high accuracy with minimal degradation as privacy increased. [112]
- **A Privacy-Preserving Federated Learning Model for Money Laundering Detection** (2025) — paired FL with DP and secure aggregation; reported accuracy up to 85.71%. [99]
- **Federated Gradient Boosting for Financial Fraud Detection** (2025) — found federated boosting strong, but vulnerable to data skew and bank join/dropout. [105]

- **Federated Learning for Privacy-Preserving Fraud Detection in Cross-Bank Systems** (2025) — showed FL comparable to centralized fraud detection while preserving privacy. [96]
- **TrustFed-HE: Trustworthy Federated Fraud Analytics for Banks with Homomorphic Secure Aggregation and Poisoning-Resilient Training** (2026) — strong attack robustness plus high AUC-PR/recall at low FPR. [98]
- **Federated Learning Integrated Fraud Detection Framework for Secure Financial Transaction** (2025) — FL + DP improved accuracy, convergence, and false-positive rate. [100]
- **Federated Learning with Differential Privacy for Fraud Detection** (2026) — performance held up under moderate heterogeneity but dropped under severe non-IID and small-client settings. [103]

Private set intersection

- **Federated K-Private Set Intersection** (2022) — privacy-preserving identifier/query matching across parties; strong information-theoretic privacy. [91]
- **Threshold Multiparty Private Set Intersection via Federated Graph Neural Networks** (2025) — thresholded matching with lower overhead and scalable performance. [92]
- **Privacy-Aware Transaction Fraud Detection With Low Communication Costs Under Joint Federated Learning** (2026) — notes that PSI can discard useful local data and motivates joint FL to reduce that loss. [93]
- **PPDFP: Privacy-Preserving Data Filtering Protocol for Large-Scale ID Alignment of Vertical Federated Learning** (2025) — PSI preprocessing to reduce runtime and communication substantially at large scale. [109]

Secure multi-party computation

- **Practical secure aggregation for privacy-preserving machine learning** (2017, foundational) — secure aggregation with dropout robustness and low overhead. [90]
- **Secure Multiparty PageRank Algorithm for Collaborative Fraud Detection** (2019) — computes graph features jointly without revealing other parties' transaction graphs. [95]
- **Privacy-enhanced clustered federated learning with secure clustering** (2026) — describes MPC as a privacy-preserving layer for FL, but with communication cost. [108]
- **Privacy-preserving and efficient verifiable federated learning scheme based on proxy re-signature** (2025) — characterizes SMPC as strong but implementation-heavy and communication-intensive. [110]
- **Zero-Knowledge Proofs in Anti-Money Laundering Multiparty Computation** (2025) — MPC plus proofs for private AML score computation and public verifiability. [94]
- **Fed-RD** and **TrustFed-HE** also use secure aggregation / SMPC-style mechanisms in financial crime detection. [98] [112]

Bottom line: for cross-institution fraud and AML detection, **FL** is the best fit for learning shared detectors, **PSI** is the best fit for privacy-preserving entity matching, and **SMPC** is the best fit for secure joint computation; the strongest real-world systems often **combine all three** because no single method solves privacy, matching, and robust collaboration at once. [91] [92] [93] [94] [98] [99] [109]

Confidence level: High — the evidence consistently supports this division of labor across multiple financial/fraud/AML studies, with recurring, well-aligned claims about strengths and trade-offs.

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